

UTKARSH PRAKASH

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PROFESSIONAL SUMMARY

Experience as a Data Scientist developing and deploying data driven solutions in the domains of finance, logistics, and manufacturing. Proven capability to develop automation pipelines, implement state of the art machine learning models, and generate measurable business outcomes. Participated and awarded in multiple global AI competitions for innovative solutions to real world problems.

SKILLS

Programming & Frameworks: Python, Scikit-Learn, TensorFlow, Hugging Face Transformers, LangChain, Streamlit

Data & Databases: SQL, Snowflake

Machine Learning & AI Concepts:

Supervised Machine Learning: Regression, Tree-based models, SVM, Ensembles, Time Series

Unsupervised Learning: Clustering, Dimensionality Reduction

Deep Learning: CNN, RNN, LSTM, GRU, Transformers

Cloud & MLOps: Azure ML, Azure Storage, Azure Functions, Azure App Services, Azure DevOps, Databricks, MLflow, Docker, Kubernetes, Git

Visualization & BI: Power BI

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Advanced Analytics and Artificial Intelligence - ABB

January 2025 to Present

- Developed text to SQL agents for finance processes with role authentication and execution confirmation steps to prevent unauthorized or unsafe queries.
- Implemented custom PDF image/table parsing to analyze competitor data; reduced competition-analysis turnaround from ~6-7 hours to minutes (~95% faster).
- Created hierarchical chunking + embeddings to maintain document structure and include source cross-references/citations in agent responses.
- Built hybrid retrieval (semantic similarity + reference/metadata filters) to enhance grounding and accuracy for finance entities.
- Developed a validation framework that encompasses prompt/agent tests and response-grounding checks.

Data Scientist, Advanced Analytics and Artificial Intelligence - ABB

March 2023 to January 2025

- Developed forecasting workflows for top-line P&L line items for 25+ divisions.
- Tuned and compared 13 models (8 multivariate, 5 univariate; temporal + regression) through cross-validated hyperparameter tuning; chose on a held-out test set with $\geq 95\%$ average accuracy.
- Built reusable model-wrapper classes to represent business rules, standardize I/O, and facilitate plug-and-play experimentation.
- Designed and deployed Azure ML pipelines along with a parallel training/execution stack, reducing end-to-end training+CV time from ~15 h to 1.5–2 h (~10 $\times$).
- Developed a Streamlit application with Azure AD-authenticated identity and row-level access; containerized and hosted on Azure App Service, handling 40–50 simultaneous users.

Consultant, Data Science and Analytics - Capgemini

April 2022 to February 2023

- Designed data matrices and reports to analyze relationships between material parameters and machine settings from factory/lab data.
- Implemented automation pipelines for daily report generation, reducing operational time for end users.
- Developed tree-based models to identify factors causing parameter threshold breaches and consumer complaints.
- Established data sources and document hierarchy for materials, plants, and routes in distribution incidents/thefts.
- Conducted route categorization and trend analysis to prevent an estimated annual loss of **USD 1.8M**

Associate Consultant, Data Science and Analytics - Capgemini

October 2020 to April 2022

- Built linear programming functions for flagging, rescheduling, and coupling operational/commercial projects.
- Optimized models to reduce processing time by **50%+**.
- Fine-tuned text matching algorithms to achieve **93% accuracy** in description standardization, improving project timeline optimization.
- Automated monthly updates for 3000+ project-related expenses; developed classification models with **95% accuracy**.
- Created retraining pipelines to capture model drift, integrate user feedback, and apply configuration updates.

Senior Analyst, Data Science and Analytics - Capgemini

July 2019 to October 2020

- Developed NLP-based product categorization application with automated model training pipelines.
- Implemented multi-tab UI to update training data, run inference, retrain, and monitor NLP models.

KEY ACHIEVEMENTS & COMPETITIONS

- **Winner**, Global Data Science Challenge 2022 - AI solution to automate evaluation in River Blindness clinical trials.
- **Second Runner-up**, Global Data Science Challenge 2021 - AI for climate change anomaly detection in subsea sensor data.
- **Runner-up in India, World Rank 11**, Global Data Science Challenge 2020 - Whale identification via image recognition.
- **XtraMile Award** (Mar 2021) - Recognized for high-value project contributions.
- **Panel Member**, Data Sangam 2020 - Guided prospective undergraduates on careers in Data Science.
- **Millennial of the Quarter** (Jun 2020) - Innovative contributions to millennial garage projects.

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Communication Engineering
SRM Institute of Science and Technology (2015 - 2019)

High School (Science Stream)
St. Michaels High School, Patna (2015)

CERTIFICATIONS

Industry Certifications:

- AWS Certified Machine Learning - Specialty
- Microsoft Certified Azure Data Scientist Associate (DP-100)
- Microsoft Certified Azure AI Engineer Associate (AI-100)
- Microsoft Certified Azure Fundamentals (AZ-900)
- Snowflake Fundamentals (Partner Training)
- Snowflake - Snowpark and Gen AI (Partner Training and Conference)

Relevant Training & Specializations:

- DeepLearning.AI TensorFlow Developer Specialization
- IBM Data Science Professional Certificate
- Machine Learning - Stanford Online
- Deep Learning Specialization - Coursera
- AWS for ML Practitioners (Partner Training)
- Data Science Math Skills - Coursera
- AI Genie Certification
- AI Cadet Certification